a Within-task recovery

Observe / repair / retry



Same episode; match retry budgets.
No persistent edit is implied.

b Retained-artifact gain

Physical reset / matched retry budget

Frozen baseline

Retained update

Same-task test

Vary the retained artifact.
Match actor, tools and supervision.

c Held-out transfer

Version A

Version B

Same held-out tasks

Declare task / object / robot shifts.
Report any target-side adaptation.

d Retention and regression

Test A before → Update on B → Re-test A

Measure earlier-task performance

Re-test old tasks after each update.
Stored knowledge is not task retention.

e Acquisition efficiency

Teaching / practice / compute budgets

Earlier procedure

Updated procedure

Same new task

Compare gain at a fixed budget,
or cost at matched task performance.

f Acquisition repeatability

Init 1 → Acquire → Test

Init 2 → Acquire → Test

Init 3 → Acquire → Test

Restart the declared acquisition scope.
Include failed histories and uncertainty.

Protected final testing

Stronger reporting condition; distinct from the minimum L4 boundary

Acquire / revise → Develop / select → Freeze artifact → Final task test

Final-test feedback must not enter artifact construction or candidate selection.

Minimum L4 test boundary: declared acquisition / evaluation separation; explicit exclusion is coded separately.
Control or report task distribution, action interface, actor, retry budget, supervision and evaluator.
Separate simulation / hardware; report task outcomes, uncertainty, interventions, resets, rescue, time and compute.